Lipid Structure Writer

Contents

1 2026.09.10 - Writing the tier-b lipid structures, and what they bought ✗ 1

1.1 The question ✗ . . . 1
1.2 What was built ✗ . . . 1
1.3 Design decisions, none of which is read off the name ✗ . . . 1
1.4 Parser coverage ✗ . . . 2
1.5 The acceptance test ✗ . . . 2
1.6 Pricing, and the two routes ✗ . . . 3
1.7 A created compound is not on the same energy zero as a cached one ✗ . . . 3
1.8 Coverage ✗ . . . 4
1.9 Figure ✗ . . . 5
1.10 The extended compound cache, and where it lives ✗ . . . 5
1.11 What remains ✗ . . . 5

✓read and checked ■read, provisional ✗agent-written, not yet read

1 2026.09.10 - Writing the tier-b lipid structures, and what they bought ✗

1.1 The question ✗

`experiments.026-metabolism-flux.external-pka-route` left one open item as the only thing that could change the reach number: 409 of the 665 unpriced yeast-GEM 9.0.2 metabolites, 206 distinct names, have a definite elemental formula and no structure in any namespace, and they are lipids whose acyl composition is written into the name. That note's estimate was an upper bound of 425 of 665 if a writer worked perfectly, with the caveat that the group-decomposition rate on freshly written lipids was unknown and 2 of the 5 pilot structures had failed decomposition.

The writer now exists: `experiments/026-metabolism-flux/scripts/lipid_structure_writer.py`, five subcommands run under three environments, plus `experiments/026-metabolism-flux/scripts/plot_lipid_structures.py` for the figure.

1.2 What was built ✗

subcommand	environment	output
`parse`	torchcell	`results/lipid_names_parsed.json`
`build`	equilibrator-ext	`results/lipid_structures.json`, `results/lipid_structures.csv`
`pkas`	pka-tools	`results/lipid_pkas.json`
`price`	equilibrator-ext	`results/lipid_priced.json`, `results/lipid_priced.csv`
`coverage`	torchcell	`results/lipid_coverage.json`

`parse` decomposes a name into a headgroup template and a list of acyl chains carrying position, carbon count and double-bond count, as pydantic models. `build` assembles the SMILES in RDKit and runs the acceptance test. `pkas` walks the MolGpKa ladder, importing `_macro_from_sites` and `_apply_proton_change` from `experiments/026-metabolism-flux/scripts/external_pka_benchmark.py` so the aggregation is the same function the 300-compound benchmark scored rather than a second implementation. `price` runs two routes and reports both. `coverage` reuses `shipped_series` from `audit_reaction_thermo_coverage.py` and `TIGHT_SPREAD_KJ` from `improve_parameter_coverage.py`; the `priceable` predicate is a closure inside `audit_reaction_thermo_coverage.main` and cannot be imported, so it is restated in three lines with a comment saying where it comes from.

A correction to the earlier note's run instructions. The environment that carries `equilibrator-assets` master with `pkas_provided_externally` is `~/miniconda3/envs/equilibrator-ext` (`0.6.0.post29+git.c79e371e`), not `/scratch/projects/torchcell-scratch/envs/equilibrator`, which is on the 0.6.0 PyPI release and has no such keyword.

1.3 Design decisions, none of which is read off the name ✗

1. **Double-bond position.** yeast-GEM writes `16:1` and never says where the double bond is. Every double bond is placed at Delta-9 counting from the carbonyl carbon, and at every third carbon after that, in the cis configuration. `build` asserts the convention rather than asserting it in prose: the free 16:1 and 18:1 acids it writes hash to `SECPZKHBENQXJG-FPLPWBNLSA-N` and `ZQPPMHVWECSIRJ-KTKRTIGZSA-N`, the InChIKeys of palmitoleic and oleic acid, and the run aborts if they do not. Group contribution reads a group vector that does not carry the

double-bond position, so the choice does not move the energy; it is recorded because the structures are published and the convention is not recoverable from the name.

2. **Stereocenters are left unspecified**, including in the coenzyme A moiety. The glycerol sn-2 carbon, the inositol ring and the (R)-3-hydroxyacyl carbon all carry stereochemistry in the real molecule and none of it is in the yeast-GEM name for the glycerolipids. Double-bond geometry IS written, since the acyl-CoA names state cis or trans and the Delta-9 convention is cis. This decision has a measured cost, in the cache-lookup section below.
3. **Protonation.** Every structure is written neutral, so the reference microspecies eQuilibrator derives from the SMILES is the neutral one. yeast-GEM declares a formula at the model's own charge state, so the acceptance test compares against the neutral formula, `H_neutral = H_gem - charge_gem` with every other element unchanged. For all 168 names that pass, `charge_gem` is 0 and the adjustment is the identity; the code carries it anyway because the tier-b set contains charged species that do not parse.
4. **Coenzyme A comes from the cache, not from a keyboard.** `build` looks up InChIKey prefix `RGJOEKWQDUBAIZ`, neutralizes the stored -4 microspecies with RDKit's `Uncharger`, asserts the result is `C21H36N7O16P3S`, and thioesterifies it with a reaction SMARTS. The 48 heavy atoms of the nucleotide are therefore not ours to get wrong.
5. **The SLIME backbone pseudo-metabolites are not molecules and no structure is attempted.** yeast-GEM's lipid pseudo-reactions carry species named `<class> backbone` whose formula is a difference formula: `triglyceride backbone` is `C3H2`, `ceramide backbone` is `HNO2`, `long-chain base backbone` is `H4N`. They are reported as unparsed with that reason rather than forced into a structure.

1.4 Parser coverage ✗

From `results/lipid_names_parsed.json`: **168 of the 206 distinct tier-b names parse, which is 347 of the 409 metabolite rows.** The 38 that do not split into exactly two reasons.

reason	n names
`slime_backbone_pseudo_metabolite`	17
`no_acyl_composition_in_name`	21

Every unparsed name, by class:

class	names
other (12)	`Apocytochrome c`, `Cytochrome c`, `Dextrin`, `Ferricytochrome b5`, `Ferrocytochrome b5`, `chitin`, `chitosan`, `heme a`, `heme o`, `mannan`, `pectin`, `tyrosyl acetate`
sphingolipid (6)	`ceramide backbone`, `inositol phosphomannosylinositol phosphoceramide backbone`, `inositol-P-ceramide backbone`, `long-chain base backbone`, `long-chain base phosphate backbone`, `mannosylinositol phosphorylceramide backbone`
glycan (4)	`(1->3)-beta-D-glucan`, `(1->6)-beta-D-glucan`, `N,N'-Chitobiosyldiphosphodolichol`, `Starch`
protein_linked (4)	`[cytochrome c]-L-lysine`, `[cytochrome c]-N6-methyl-L-lysine`, `[protein]-L-lysine`, `[protein]-N(6)-acetyl-L-lysine`
phosphatidylinositol_PI (3)	`1-phosphatidyl-1D-myo-inositol backbone`, `sn-2-acyl-1-lysophosphatidylinositol backbone`, the GPI anchor `6-[6-(2-aminoethylphosphoryl)-alpha-mannosyl-(1->6)-...]-O-acyl-1-phos`
triacylglycerol_TAG (2)	`diglyceride backbone`, `triglyceride backbone`
one each (7)	`cardiolipin backbone`, `ergosterol ester backbone`, `phosphatidate backbone`, `phosphatidyl-L-serine backbone`, `phosphatidylcholine backbone`, `phosphatidylethanolamine backbone`, `phosphatidylglycerol backbone`

The sphingolipids are all backbones. The reach analysis counted 12 sphingolipid metabolites in tier b, and every one of them is a SLIME pseudo-metabolite: yeast-GEM 9.0.2 carries no ceramide, inositol phosphoceramide or mannosylated inositol phosphoceramide with an acyl composition in its tier-b name, so there was nothing for a ceramide template to build. The template set covers glycerophospholipids (PA, PE, PMME, PDME, PC, PS, PG, PGP, PI and its 3-, 4-, 3,5- and 4,5-phosphates, DGPP, CDP-DAG), triacyl-, diacyl- and monoacylglycerols, cardiolipin and monolysocardiolipin, lysophosphatidylinositol, acylglycerone phosphate, free fatty acyl chains and acyl-CoAs. PS and DAG have templates and no tier-b name uses them; they exist because the same parser is run over all 2,806 yeast-GEM metabolites for the figure's comparison population, where it recognizes 287 distinct lipid names.

1.5 The acceptance test ✗

168 of 168 written structures reproduce yeast-GEM's own formula exactly (`results/lipid_structures.json`, `n_formula_pass: 168`, `n_formula_fail: 0`, covering 347 metabolite rows). There are no failures to list. By template: CL 60, TAG 18, ACYLCOA 9, PI35P2 8, DGPP 7, PI45P2 7, PI4P 7, PDME 7, PMME 7, PI3P 6, FFA 6, PE 6, PGP

5, CDPDAG 5, MAG1 3, PC 2, PG 2, PI 1, ACYLDHAP 1, LPI2 1.

A zero-failure acceptance test is weak evidence on its own, so two independent checks were run. The Delta-9 witnesses above are one. The other is the cache: 120 of the 168 written InChIKeys resolve a compound eQuilibrator already holds, which is an external structure nobody here wrote. Our phosphatidylcholine (1-16:0, 2-18:1) hashes to `WTJKGGKOPKCXLL-MRCUWXFGSA-N` and POPC in the cache to `WTJKGGKOPKCXLL-VQVTYTSYSA-N`, the same constitution written with and without the sn-2 center.

1.6 Pricing, and the two routes ✗

`price` runs two routes per structure at pH 7.2, I = 0.25 M, T = 303.15 K, pMg 3.0.

Route 1, cache lookup. The audit could never try this: it had no structure to key on. A written structure gives one. The search runs twice, first on the skeleton and stereochemistry blocks (the fix `pathway_thermo.py` needed, since the cache stores the major microspecies and the last block records the protonation state), then on the skeleton alone, because design decision 2 leaves stereocenters unspecified while the cache's entries often carry them. Both passes require every element but hydrogen to match the written formula, so a hit cannot be a different molecule. Result: **120 of 168, split 19 on stereo plus skeleton and 101 on skeleton alone.**

Route 2, external pKa. MolGpKa ladders for all 168 structures in 74 s (`results/lipid_pkas.json`), then `LocalCompoundCache(pkas_provided_externally=True)` on assets master. 148 compounds were created, 20 were not because their ladder is empty, and **148 of 148 created compounds passed group decomposition.**

That last number answers the earlier note's open question. Group decomposition was flagged as an unmeasured ceiling after 2 of 5 pilot structures failed it; on written lipids the failure rate is 0 of 148. The two pilot failures, ferroheme b and methionyl acetate, are an iron porphyrin and an ester of an amino acid, not lipids, so they were never evidence about this class.

All 168 names are priced, 120 by cache lookup and 48 by the external route, covering 347 metabolite rows (`results/lipid_priced.json`).

The 20 empty ladders are 18 triglycerides and 2 phosphatidylcholines, and they are the shape the interface cannot express: a triacylglycerol has no ionizable group and a phosphatidylcholine zwitterion has none between pH 0 and 14, so the correct pKa list is empty, and `local_compound_cache.py:128` guards on truthiness and raises `MissingPKasError` instead of creating a single-microspecies compound. All 20 were recovered by the cache lookup, so the interface limitation costs nothing here, but it is still a real defect and it will bite any compound that is genuinely absent and genuinely non-ionizable.

Of the 48 that only the external route reached, 26 are cardiolipins, 6 DGPP, 6 PDME, 6 PMME, 3 CDP-DAG and 1 acyl-DHAP. None of their ladders hit the rung budget.

1.7 A created compound is not on the same energy zero as a cached one ✗

This is the result that changes how the numbers should be used, and it was not anticipated.

For 90 names both routes produced a number. Their `dGf'0` values differ by 1930 kJ/mol on average, up to 3023. That is not a matching error: the InChIKey skeleton and every heavy atom agree, so the two are the same chemical. Restricting to the 7 pairs whose reference microspecies carries the same number of protons isolates the drawing convention from the protonation state, and there the gap is almost a constant:

name	formula	cache minus external (kJ/mol)
phosphatidylethanolamine (1-16:0, 2-16:1)	C37H72NO8P	-776.11
phosphatidylethanolamine (1-16:0, 2-18:1)	C39H76NO8P	-776.11
phosphatidylethanolamine (1-16:1, 2-18:1)	C39H74NO8P	-776.11
phosphatidylethanolamine (1-18:0, 2-16:1)	C39H76NO8P	-776.11
phosphatidylethanolamine (1-18:0, 2-18:1)	C41H80NO8P	-776.10
phosphatidyl-N-methylethanolamine (1-16:0, 2-16:1)	C40H78NO8P	-779.22
phosphatidyl-N,N-dimethylethanolamine (1-16:0, 2-16:1)	C41H80NO8P	-782.84

Taking phosphatidylethanolamine (1-16:0, 2-16:1) apart: the cache compound is id 57341, SMILES `...COP([O-])(=O)OCC[NH3+]` atom bag `{C: 37, H: 72, N: 1, O: 8, P: 1}`, untransformed -1087.72 kJ/mol, Legendre transform +3066.82, `dGf'0` 1979.09. The created compound is id 694430, SMILES `...COP(=O)(O)OCCN`, the same atom bag, untransformed -311.47, transform +3066.68, `dGf'0` 2755.20. **The transforms agree to 0.14 kJ/mol and the entire gap is in the untransformed term**, which is the group-contribution estimate for whichever microspecies the SMILES draws. Plausible mechanism, unverified: eQuilibrator's component contribution was trained on ChemAxon's major microspecies, so the group vector for a phosphate anion beside an ammonium and the group vector for a phosphate acid beside an amine are both in range but are not required to agree, and drawing the un-ionized protomer puts the compound outside the convention.

Two consequences.

- **Where a cache compound exists it is the value reported**, because the recomputed table the flux layer reads is built from cache compounds. That is what `route: cache_lookup` means in `results/lipid_priced.csv`, and it is 120 of 168 names, 289 of 347 metabolite rows.
- **The 48 external-route values carry an offset of unknown size.** It is 777 kJ/mol on the aminophospholipids where it can be measured, and it has not been measured on a cardiolipin, a DGPP or a CDP-DAG, which is what those 48 are. They are reported and counted, and the coverage arithmetic below gives the cache-route-only number beside the full one so a claim can pick the conservative one.

This also retires a reading of the earlier pilot. That note put the written phosphatidylethanolamine (1-16:1, 2-16:1) at 1967.93 kJ/mol beside the cache-priced phosphatidylethanolamine (1-18:1, 2-16:1) at 2413.32 and called them the same family. They are not on one zero, and the difference between two such numbers is not a chemistry difference.

1.8 Coverage ✗

From `results/lipid_coverage.json`, over yeast-GEM 9.0.2's 4,131 reactions and 2,806 metabolites.

347 of the 665 unpriced metabolites are now priced, 52.2 percent, of which 289 come through the cache lookup and 58 through the external route. By class: PI 131, CL 60, TAG 51, PE 50, PA 23, acyl-CoA 9, fatty acyl chain 8, PC 8, PG 7.

route	reactions with every participant priced	fraction
eQuilibrator recomputation, before	2741	66.35 %
plus written lipids, cache route only	3257	78.84 %
plus written lipids, both routes	3564	86.27 %
yeast-GEM shipped metabolite column	3505	84.85 %
union of shipped and recomputed, before	3716	89.95 %
plus written lipids	3716	89.95 %
recommended set (union plus tight surrogates), before	3730	90.29 %
plus written lipids	3732	90.34 %
recommended, restricted to the 2,845 flux-carrying reactions, before	2685	94.38 %
plus written lipids	2685	94.38 %

The gain is 823 reactions on the eQuilibrator route and 0 on the union. Read those two rows together, because the second is the one that governs what to claim. yeast-GEM's shipped metabolite column already carries values for these lipids, matched by metabolite name out of yETFL and ModelSEED, so at the union level the writer adds nothing: 3716 before and 3716 after. The recommended set gains 2 reactions, and the flux-carrying restriction that the parameter-coverage report quotes at 94.38 percent does not move at all.

What the writer buys is **method consistency, not coverage**. Before it, one self-consistent eQuilibrator recomputation covered 2,741 reactions, 764 fewer than the shipped column's 3,505, and the shipped column is three different methods behind one column (yETFL by name, ModelSEED by KEGG id, dG-predictor by row position; see `audit_reaction_thermo_coverage.py`'s `SHIPPED_PROVENANCE`). After it, the eQuilibrator route covers 3,564, which is 59 more than the shipped column, and every one of those values comes from the same group-contribution method with an uncertainty attached. That is the result worth quoting, and the 90.0 percent union is unchanged.

1.9 Figure ✗

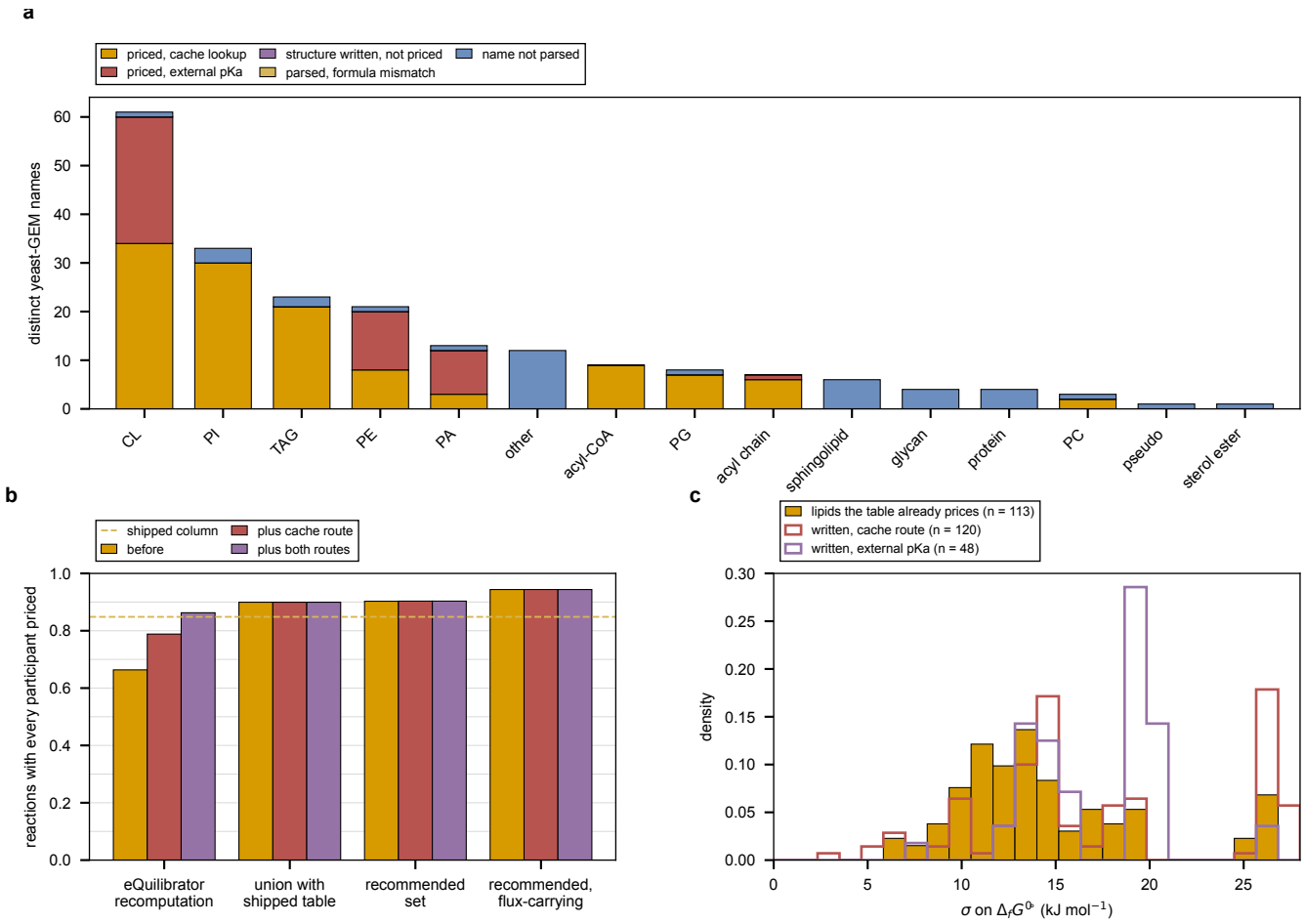


Figure 1. What the lipid structure writer reached, and what it bought. a, The 206 distinct tier-b names by yeast-GEM chemical class, each bar partitioned into the five mutually exclusive stages of the pipeline. No name is lost to a formula mismatch and none to a written structure that failed to price, so those two segments are empty everywhere; the whole loss is the 38 names the parser refuses, which are the 17 SLIME backbone pseudo-metabolites and 21 names with no acyl composition (glycans, hemes, cytochromes, protein-linked lysines, a GPI anchor). **b**, Reactions of 4,131 with every participant priced, before the written lipids, with the 120 names that resolve an existing cache compound, and with all 168. The eQuilibrator recomputation moves from 66.35 to 86.27 percent and passes the shipped metabolite column's 84.85 percent (dashed). The union with the shipped table does not move at all, because that table already prices these lipids. **c**, Uncertainty on the formation energy, one value per distinct name: the 113 lipid names the recomputed table already prices, the 120 written names priced by cache lookup, and the 48 priced through MolGpKa pKas and a created compound. Medians 13.19, 14.79 and 18.92 kJ/mol. Drawn by `experiments/026-metabolism-flux/scripts/plot_lipid_structures.py` from `results/lipid_names_parsed.json`, `results/lipid_structures.json`, `results/lipid_priced.json` and `results/lipid_coverage.json`.

1.10 The extended compound cache, and where it lives ✗

Compounds created here are written to `/scratch/projects/torchcell-scratch/tmp/lipid_ccache.sqlite`, a copy of the shipped cache made before the run. **The shipped cache at `~/.cache/equilibrator/compounds.sqlite` is never opened for writing**, and `price` refuses to run if `--cache` resolves to it, because a compound created here would otherwise be indistinguishable from a MetaNetX one on the next read. The copy holds the 148 created compounds and is reusable: point `price --cache` at it, or open it with `LocalCompoundCache.load_cache`. It sits under `DATA_ROOT/tmp` rather than in the agent job directory so that it survives the session, next to the pilot's `pka_local_ccache.sqlite`.

1.11 What remains ✗

- **The 777 kJ/mol convention offset is measured on aminophospholipids only.** Closing it means writing each structure as its major microspecies at pH 7.2 rather than as the neutral parent, which the MolGpKa ladder already contains enough information to do. Until that is done the 48 external-route values should not be mixed into a table with cache values, and `lipid_coverage.json` reports the cache-route-only counts for that reason.
- **The compartment-pH mismatch.** The new values are all at pH 7.2, while `metabolite_dgf_prime.parquet` transforms each metabolite at its own compartment pH: 5.0 extracellular and in the cell envelope, 6.2 in the vacuole, 6.6 in the Golgi, 7.0 in the peroxisome, 7.1 in the endoplasmic reticulum, 7.8 in the mitochondrion, 7.2 in the cytoplasm, nucleus and lipid particle. Only the reaction COUNTS in the table above are comparable; before any of these numbers enters the flux layer they need re-transforming per compartment, which is a call to `Compound.transform` and not a re-derivation.

- **Magnesium is still unsuppliable** through `specified_pkas`, so the 48 created compounds carry an empty magnesium dissociation list. Measured cost elsewhere: 0.32 kJ/mol for UDP, 1.10 kJ/mol for ADP.
- **38 names remain unwritable and 318 of the 665 metabolites remain unpriced.** The 17 backbones are pseudo-metabolites and no structure should exist for them; the 21 others are polymers, metalloproteins and protein-linked residues, which is tier c's ceiling arriving under a different label.
- **The empty-pKa interface defect** (`local_compound_cache.py:128` guards on truthiness) is worth reporting upstream. It cost nothing here only because every affected compound happened to be in the cache already.